

1) Classification Algorithm

a) Age <30 and Govt satisfaction less ,urban

In Application:

MAKE A PREDICTION

Age

19.0 yrs

Govt. satisfaction

5.30 /10

Urban voter

Yes

No

Yes

Predict

PREDICTED PREFERENCE

Undecided

Party A

1.8%

Party T

46.8%

Party C

0.2%

Undecided

51.3%

ChatGPT:

Percentages

- **Party A:** $0.202/11.403 \times 100 \approx 1.77\%$
- **Party T:** $5.312/11.403 \times 100 \approx 46.58\%$
- **Party C:** $0.01795/11.403 \times 100 \approx 0.16\%$
- **Undecided:** $5.871/11.403 \times 100 \approx 51.49\%$

Total = 100.00%

So the result is:

Party	Percentage 
Party A	1.77%
Party T	46.58%
Party C	0.16%
Undecided	51.49%

The result is “Undecided” in both application and in chatGPT.

b) Age >50 and Govt satisfaction less ,non urban
validate the percentage

In Application:

MAKE A PREDICTION

Age

70.0 yrs

Govt. satisfaction

0 /10

Urban voter

Yes

No

No

Predict

PREDICTED PREFERENCE

Party C

Party A

0.0%

Party T

0.5%

Party C

99.4%

Undecided

0.0%

ChatGPT:

Result

Party C: ~99.42%

Party T: ~0.55%

Undecided: ~0.03%

Party A: ~0.000003%

So the model gives an overwhelming probability to **Party C** (about 99.4%).

Result: Party C in both application and in chatGPT.

2. Linear Regression:

a. QA Salary

Application:

MAKE A PREDICTION

Experience 8.00 yrs

Skill rating 8.30 /10

Predict

TOTAL PACKAGE 22.0 LPA

Base salary – the starting value, before any feature counts
3.10 LPA

Experience
 $0.966 \text{ LPA per year} \times 8.00 \text{ yrs}$
= +7.73 LPA

Skill rating
 $1.34 \text{ LPA per point} \times 8.30 /10$
= +11.1 LPA

TOTAL 22.0 LPA

Gemini:

Based on the formula components provided, your total calculated salary is **21.95 LPA**.

The breakdown of the calculation is as follows:

- **Base Salary: 3.10 LPA**
- **Experience Component:** 8.00 years × 0.966 LPA/year = **7.73 LPA** (rounded from 7.728 LPA)
- **Skill Component:** 8.30 points × 1.34 LPA/point = **11.12 LPA** (rounded from 11.122 LPA)

$$\text{Total Salary} = 3.10 + 7.728 + 11.122 = \mathbf{21.95 \text{ LPA}}$$

Rounded to 22 LPA

Result : Salary 22 LPA in both application and in Gemini.

b) Food Delivery time:

Application:

MAKE A PREDICTION

Distance

12.0 km

Traffic

8.00 /10

Predict

TOTAL TIME

61.6 min

Base time – the starting value, before any feature counts

7.56 min

Distance

2.40 min per km × 12.0 km

= +28.8 min

Traffic

3.16 min per point × 8.00 /10

= +25.3 min

TOTAL

61.6 min

Gemini:

The total calculated time is **61.66 minutes** (or **1 hour, 1 minute, and 40 seconds**).

Here is the breakdown of the calculation:

Feature	Calculation / Value	Result
Base Time	Starting value	+7.56 min
Distance	$2.40 \text{ min/km} \times 12.0 \text{ km}$	+28.80 min
Traffic	Value provided in prompt	+25.30 min
Total Time	$7.56 + 28.8 + 25.3$	61.66 min

Result: Total time is 61.66 mins